

⚡ 1-HOUR MASTER REFERENCE · GEN AI ARCHITECT

The GenAI Stack

Everything a Generative AI Architect needs — from LLMs to deployment, RAG to agents, vector stores to Python libraries.

9
TOPICS COVERED

40+
LLM MODELS

12+
VECTOR DBS

30+
PYTHON LIBS

6
CHATBOT TYPES

01

Large Language Models (LLMs)

GPT-4o / GPT-4 TurboOPENAI

Multimodal flag­ship. Han­dles text, image, audio. 128K context win­dow. Top bench­mark per­for­mance. Pow­ers ChatGPT Enter­prise.

Multimodal128K ctxAPI

o1 / o3 / o4-miniOPENAI

Reasoning-first mod­els. Extended chain-of-thought. Best for math, coding, logic. Slower but dram­atically more accu­rate on hard tasks.

Claude 3.5 / 4 Sonnet

ANTHROPIC

Superior instruction following. 200K context. Excellent for long docs, coding agents, safe deployments. Opus for most capable tasks.

- 200K ctx
- Safety
- Coding

Gemini 2.0 / Ultra

GOOGLE

1M token context (Flash). Native multimodal. Deep Google Search integration. Best for long-context retrieval and document analysis.

- 1M ctx
- Google
- Multimodal

Llama 3.1 / 3.3

META · OSS

Open source. 8B to 405B params. Self-hostable. Fine-tuning friendly. Best OSS choice for on-premise and regulated environments.

- Open Source
- Self-host
- Fine-tune

Mistral / Mixtral

MISTRAL · OSS

MoE architecture (Mixtral 8x7B). Efficient, fast, multilingual. Mistral Large rivals GPT-4 at lower cost. Great for European deployments.

- MoE
- Multilingual
- OSS

Command R+ / Aya

COHERE

Optimized for RAG and enterprise search. Built-in grounding, citation. Aya for 100+ language support. Best for production RAG pipelines.

- RAG-native
- Citations
- Enterprise

Falcon / Yi / Qwen

VARIOUS · OSS

Falcon (TII), Yi (01.AI), Qwen (Alibaba). Strong open-weight competitors. Qwen2.5 excels in coding; Yi in multilingual tasks.

- Coding
- Multilingual
- OSS

DeepSeek R1 / V3

DEEPSEEK

State-of-art reasoning at very low cost. R1 rivals o1 on MATH/coding. V3 is a 671B MoE. Game-changer for cost-efficient deployments.

- Reasoning
- Low-cost
- MoE

Phi-4 / Phi-3 (Small)

MICROSOFT

SLMs punching above weight. 3.8B-14B params. Ideal for edge, mobile, and constrained environments. Excellent at reasoning per param.

- SLM
- Edge
- Efficient

Grok 3

XAI

X (Twitter) data advantage. Real-time web access. Strong at current events.

Embedding Models

VARIOUS

text-embedding-3-large (OpenAI), embed-v3 (Cohere), BGE-M3, E5-large,

Competitive reasoning. Available via xAI API.

Real-timeWebAPI

nemonic-embed. Essential for RAG semantic search pipelines.

EmbeddingsRAGSemantic

02

 RAG — Retrieval-Augmented Generation

Naive RAG

BASIC

Classic pipeline: chunk → embed → store → retrieve top-k → prompt. Simple, fast to implement. Suffers with complex queries and context boundaries.

BeginnerSimpleTop-K

Advanced RAG

INTERMEDIATE

Query rewriting, HyDE (Hypothetical Doc Embeddings), re-ranking with cross-encoders, sentence-window retrieval, parent-child chunking.

Re-rankingHyDERewrite

Modular RAG

ADVANCED

Plug-and-play retrieval modules: fusion retrieval, iterative RAG, FLARE (Forward-Looking AR). Composes retrieval strategies dynamically.

FLAREFusionIterative

GraphRAG

GRAPH

Microsoft's knowledge-graph-based RAG. Extracts entities/relationships, builds community summaries. Superior for multi-hop reasoning across documents.

Knowledge GraphNeo4jMulti-hop

Agentic RAG

AGENTS

LLM decides when/what to retrieve. Router agents, sub-question decomposition, self-RAG with critique loops. LlamaIndex's agentic flows.

RouterSelf-RAGDecompose

Multimodal RAG

MULTI

Retrieves text + images + tables + charts. Uses CLIP/ColPali for image embedding. Extract from PDFs with vision models before embedding.

ImagesCLIPColPali

Hybrid Search RAG

SEARCH

BM25 keyword + dense vector search combined via RRF (Reciprocal Rank Fusion). Best accuracy in production – keyword catches exact matches vectors miss.

- BM25
- Dense
- RRF

RAG Evaluation

EVAL

RAGAS framework: faithfulness, answer relevancy, context recall, context precision. Also TruLens, DeepEval. Must-do before production.

- RAGAS
- TruLens
- DeepEval

03



Agentic AI Workflows

ReAct Agent

PATTERN

Reason → Act → Observe loop. LLM reasons about which tool to call, executes it, observes output, repeats. Foundation of most agents.

- Reasoning
- Tools
- Loop

Plan-and-Execute

PATTERN

Planner LLM creates task list; executor LLM runs each step. Decoupled planning = better long-horizon task execution. Used in AutoGPT-style systems.

- Planning
- Long-horizon
- Decompose

Multi-Agent (MAS)

PATTERN

Orchestrator + specialist agents. Researcher, Coder, Critic, Writer roles. LangGraph, CrewAI, AutoGen. Parallel work, reduces single-context limits.

- CrewAI
- LangGraph
- AutoGen

Tool Use / Function Calling

CORE

LLM selects and calls structured tools: web search, code exec, database queries, APIs. OpenAI, Anthropic, Gemini all support native function calling.

- Tools
- APIs
- Code Exec

Reflection / Self-Critique

PATTERN

Agent evaluates its own output, critiques it, and refines. Reflexion paper pattern. Dramatically improves output quality on complex tasks.

Computer Use Agent

FRONTIER

Agents control browser, desktop, and APIs visually. Claude Computer Use, Anthropic's Operator, GPT-4o vision loops. True digital worker paradigm.

Reflexion

Self-eval

Refine

Browser

GUI

Operator

Memory Systems

STATE

Short-term (in-context), long-term (vector store), episodic (conversation history), semantic (knowledge graphs). Mem0, Zep, LangMem for persistence.

Mem0

Zep

Long-term

Human-in-the-Loop

SAFETY

Checkpoints where humans approve actions. LangGraph's interrupt/resume. Essential for high-stakes agents: legal review, financial decisions.

Approval

Safety

LangGraph

04



Vector Databases — Which & Why

Pinecone

MANAGED

Fully managed, serverless. No infra management. Namespaces for multi-tenancy. Why: fastest time-to-production, built-in metadata filtering. Best for startups.

Serverless

Managed

Scale

Qdrant

OSS + CLOUD

Rust-based, blazing fast. Named vectors, payload filtering, sparse+dense (SPLADE). Why: highest performance, best for filtering-heavy queries.

Rust

Fast

SPLADE

pgvector (Postgres)

SQL + VECTOR

Weaviate

OSS + CLOUD

GraphQL API, hybrid search out-of-box, multi-tenancy, modules for text2vec/CLIP. Why: best for complex schemas and hybrid BM25+vector search.

GraphQL

Hybrid

Modules

Chroma

OSS + DEV

Embedded Python-native DB. Zero infra for prototyping. Why: instant local setup for RAG dev, LangChain/LlamaIndex default. Not for production scale.

Local

Dev

LangChain

Milvus / Zilliz

ENTERPRISE

Vector extension for PostgreSQL. Why: use if you already have Postgres. Combine ACID transactions with vector search. Supabase, Neon support it.

ACID

SQL

Supabase

Cloud-native, billion-scale vectors. HNSW, IVF, DiskANN indexes. Why: enterprise-grade for massive scale; Zilliz is managed Milvus.

Billion-scale

HNSW

Enterprise

Redis (VSS)

IN-MEMORY

Redis Vector Similarity Search. Why: sub-millisecond latency when data fits in memory. Great for real-time recommendation and session-level RAG.

In-memory

Real-time

Low-latency

FAISS / Annoy

LIBRARY

Meta's FAISS: CPU/GPU ANN library. Not a DB, but a search kernel. Use when you control infra. Annoy: memory-mapped, great for static corpora.

Library

ANN

GPU

Azure AI Search

CLOUD

Microsoft's managed search. Hybrid (semantic + keyword), integrated Azure OpenAI. Why: best choice for Microsoft-stack enterprise RAG.

Azure

Hybrid

Enterprise

Elasticsearch (ELSER)

ENTERPRISE

Elastic's sparse embedding model. If you already run ES, add vector search. Why: operational familiarity; keep existing search infra.

Sparse

ELSER

Enterprise

05

 Knowledge Base Design

Document Ingestion

LAYER 1

PDF/DOCX/HTML/Markdown loading. Unstructured.io for complex layouts. LlamaParse for PDF tables. Docling (IBM). OCR via Tesseract or Azure Form Recognizer.

LlamaParse

Unstructured

OCR

Chunking Strategies

LAYER 2

Fixed-size, sentence, paragraph, semantic, recursive, parent-child (small-to-big). Chunk size matters: 256-512 tokens for retrieval, 1K-2K for context.

Semantic

Parent-child

Overlap

Metadata & Filtering

LAYER 3

Attach source, date, doc_type, author, section. Self-query retrieval converts natural language filters to metadata queries. Critical for access control.

Metadata

ACL

Self-query

Indexing Pipeline

LAYER 4

Embed → upsert to vector DB. Use sparse+dense for hybrid. Add full-text index alongside. Keep sync with source via webhook or scheduled job.

Embed

Upsert

Sync

Knowledge Graph

ADVANCED

Neo4j, Nebula, or Amazon Neptune. Entities + relationships. LlamaIndex's KnowledgeGraph index. GraphRAG uses this for community detection.

Neo4j

Entities

GraphRAG

Freshness & Versioning

OPS

Timestamp metadata, delta ingestion, soft-delete stale docs. Track embedding model version – re-index when switching models.

Freshness

Delta

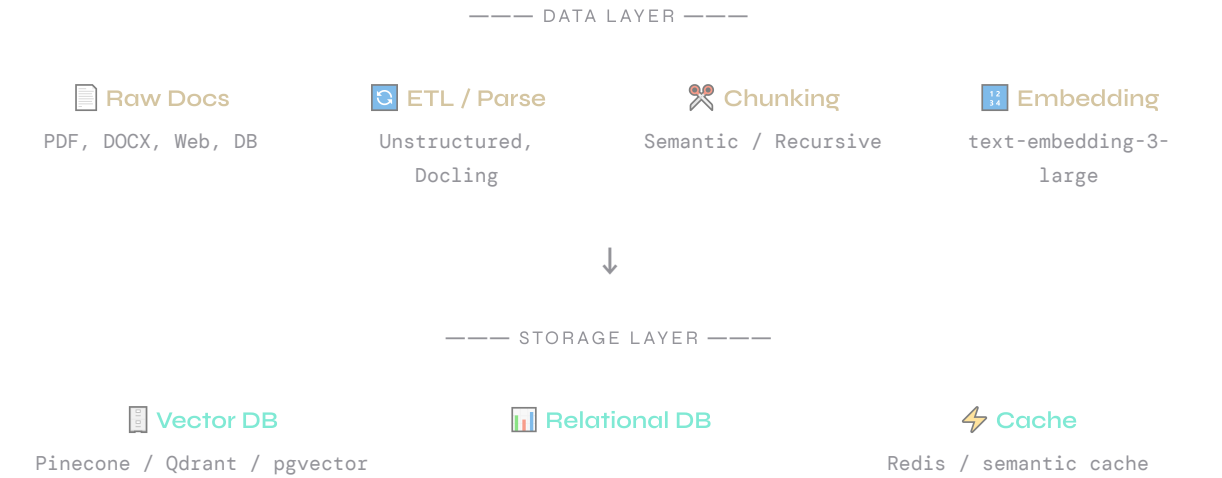
Re-index

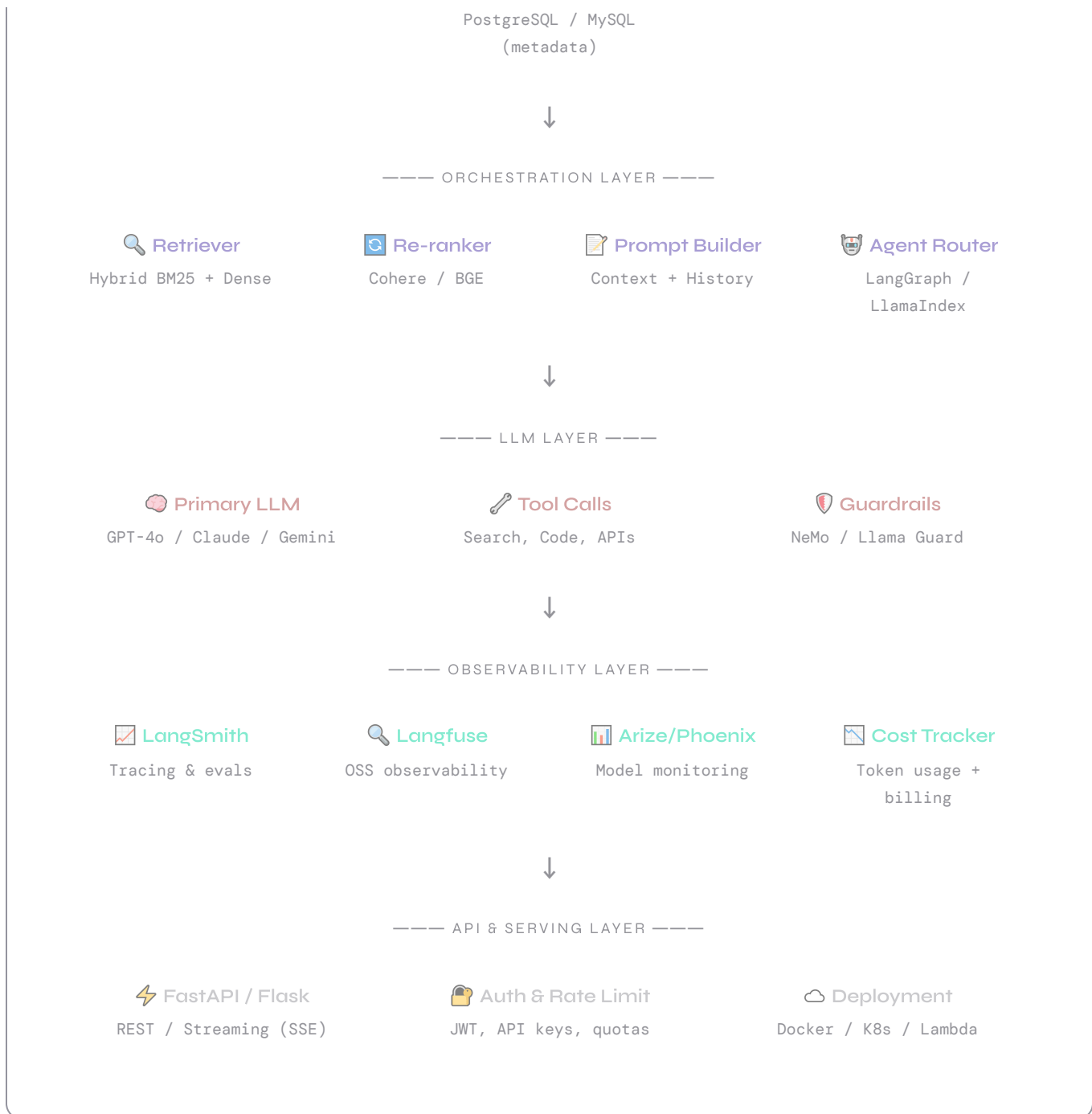
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GenAI Background Architecture

Full-stack GenAI application architecture – from data ingestion to end-user delivery.





07



Types of Chatbots

Rule-Based

GEN 1

Decision tree / if-else flows.
Dialogflow CX, Amazon Lex intents.
Deterministic, auditable. Works well

NLP / Intent-Based

GEN 2

BERT-based NLU classifies intent + extracts entities. Rasa, Botpress.
Better at variation handling but still intent-limited.

for narrow FAQ or form-filling. No LLM needed.

- Dialogflow
- Deterministic
- FAQ

- Rasa
- NLU
- Entities

RAG Chatbot

GEN 3

LLM + knowledge base retrieval. Answers from enterprise documents. Handles novel questions without fine-tuning. Most common enterprise pattern today.

- Knowledge
- Grounded
- Citations

Generative (Open-domain)

GEN 3

Pure LLM generation – ChatGPT, Claude.ai style. Creative, flexible. Needs guardrails to prevent hallucinations in business context.

- Creative
- Open-domain
- ChatGPT

Agentic Chatbot

GEN 4

Takes actions: books tickets, runs code, calls APIs, sends emails. LLM + tools + memory. Copilot Studio, Agentforce. The future of enterprise automation.

- Actions
- Tools
- Automation

Voice / Multimodal

FRONTIER

STT (Whisper) + LLM + TTS (ElevenLabs/Azure). Real-time voice assistants. GPT-4o native audio. Multimodal adds image/video understanding.

- Whisper
- Voice
- Real-time

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GenAI Architect Learning Roadmap

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PHASE 1 · WEEKS 1-3

Python & ML Foundations

Solid Python: OOP, async/await, type hints, virtual envs. NumPy, Pandas for data wrangling. Basics of Transformers architecture – attention heads, positional encoding. Understand tokenization (tiktoken, HuggingFace tokenizers).

- Python async
- Transformers paper
- HuggingFace
- Tokenizers
- Attention



PHASE 2 · WEEKS 4-6

LLM APIs & Prompt Engineering

OpenAI, Anthropic, Gemini SDKs. System prompts, few-shot, chain-of-thought, structured output (JSON mode). LiteLLM for provider abstraction. Prompt versioning with PromptLayer. Learn to measure token costs.

OpenAI SDK

Prompt CoT

LiteLLM

JSON mode

PromptLayer



PHASE 3 · WEEKS 7-10

RAG Systems & Vector Stores

Build end-to-end RAG: load → chunk → embed → store → retrieve → generate. Master LangChain and LlamaIndex. Experiment with Chroma locally, then Pinecone/Qdrant. Implement hybrid search. Evaluate with RAGAS.

LangChain

LlamaIndex

Chroma

Pinecone

RAGAS

HyDE



PHASE 4 · WEEKS 11-14

Agentic Systems & Orchestration

Build ReAct agents with tool use. LangGraph for stateful multi-agent flows. CrewAI for role-based agents. Implement memory with Mem0. Add human-in-the-loop approval. Observe with LangSmith.

LangGraph

CrewAI

Mem0

Tool calling

LangSmith



PHASE 5 · WEEKS 15-18

Fine-Tuning & Open-Source Models

LoRA/QLoRA with PEFT library. Supervised fine-tuning on instruction datasets. Run Llama with Ollama locally. Use Unsloth for 2x faster training. RLHF basics with TRL library.

PEFT / LoRA

Unsloth

TRL / RLHF

Ollama

Axolotl



PHASE 6 · WEEKS 19-22

Production & MLOps

FastAPI async endpoints with streaming (SSE). Docker containerization. Rate limiting, auth, cost controls. Langfuse/Arize for observability. CI/CD for prompt regression. Deploy to AWS/GCP/Azure.

- FastAPI
- Docker
- Langfuse
- Guardrails
- CI/CD

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PHASE 7 · ONGOING

Expert Specialization

Deep dive into GraphRAG, multimodal systems, voice pipelines, or enterprise governance. Contribute to OSS. Follow Arxiv papers weekly (ICLR, NeurIPS). Build a public portfolio project.

- GraphRAG
- Multimodal
- Papers
- OSS Contrib
- Portfolio

09



Essential Python Libraries

LIBRARY	CATEGORY	PURPOSE & WHY USE IT
openai	LLM	Official OpenAI SDK. Chat, embeddings, images, assistants, function calling.
anthropic	LLM	Anthropic Claude SDK. Messages API, streaming, tool use, vision.
litellm	LLM	Unified interface for 100+ LLM providers. Drop-in OpenAI replacement.
langchain	ORCHESTRATION	Chains, agents, retrievers. Huge ecosystem. Core RAG framework.
langgraph	ORCHESTRATION	Stateful, graph-based agent workflows. Cycles, branching, human-in-loop.

LIBRARY	CATEGORY	PURPOSE & WHY USE IT
llama-index	ORCHESTRATION	Data framework for LLMs. Best-in-class for complex RAG and indexing.
haystack	ORCHESTRATION	Deepset's pipeline framework. Production-grade RAG pipelines.
crewai	AGENTS	Role-based multi-agent orchestration. Crews, Tasks, Agents abstraction.
autogen	AGENTS	Microsoft multi-agent conversations. Code execution, tool use, group chat.

GenAI.Ref	LLMS	RAG	AGENTIC	VECTOR DB	KNOWLEDGE BASE	ARCHITECTURE	CHATB
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chromadb	VECTOR DB	Local embedded vector store. Instant RAG prototyping in Python.
pinecone-client	VECTOR DB	Pinecone managed vector DB SDK. Upsert, query, namespaces.
qdrant-client	VECTOR DB	Qdrant Python client. Fast ANN search with payload filtering.
weaviate-client	VECTOR DB	Weaviate Python SDK. GraphQL queries, hybrid search.
pgvector	VECTOR DB	Python adapter for pgvector in Postgres.
faiss-cpu/gpu	VECTOR DB	Meta ANN library. IVF, HNSW indexes for large-scale local search.
transformers	ML / NLP	HuggingFace transformers. Load BERT, Llama, Mistral, any OSS model.
sentence-transformers	ML / NLP	Semantic embeddings. SBERT, BGE, E5 models for RAG.
peft	ML / NLP	Parameter-Efficient Fine-Tuning: LoRA, QLoRA, prefix tuning.

LIBRARY	CATEGORY	PURPOSE & WHY USE IT
trl	ML / NLP	RLHF training: SFT, DPO, PPO. HuggingFace's RL for LLMs.
unsloth	ML / NLP	2x faster, 60% less VRAM for Llama/Mistral fine-tuning.
tiktoken	ML / NLP	OpenAI's fast tokenizer. Count tokens, manage context windows.
ragas	EVALUATION	RAG evaluation: faithfulness, context precision, answer relevancy scores.
deepeval	EVALUATION	LLM evaluation framework. Custom metrics, regression testing.
langsmith	EVALUATION	LangChain's tracing & eval platform. Debug, monitor production.
guardrails-ai	SAFETY	Input/output validation. Schema enforcement, content filtering.
fastapi	SERVING	Async API framework. SSE streaming, WebSocket for chatbots.
pydantic	SERVING	Data validation + structured output parsing from LLMs.
instructor	SERVING	Structured LLM outputs via Pydantic. Retry logic built-in.
unstructured	INGESTION	Parse PDFs, DOCX, HTML, images into clean text for RAG.
pypdf / pdfplumber	INGESTION	PDF text + table extraction. pdfplumber for structured tables.
streamlit / gradio	UI	Rapid chatbot UI prototyping. Demo apps in 10 lines of Python.

LIBRARY	CATEGORY	PURPOSE & WHY USE IT
langfuse	OBSERVABILITY	OSS LLM observability. Traces, cost tracking, prompt mgmt.
ollama (python)	LOCAL LLM	Run Llama, Mistral, Phi locally. Zero-cloud RAG development.

GenAI Architect Reference · Built for the 1-hour session · Cover LLMs → RAG → Agents → Vector DBs → Architecture → Libraries